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Annotation Reliability in Malaysian Malay-English Code-Mixed Sentiment Analysis

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Abstract

Sentiment analysis of Malaysian Malay-English code-mixed text depends on labels that reflect how bilingual users express opinions. Yet the resources reviewed here provide limited evidence that their labels are reliable. This review examines the linguistic difficulties behind annotation, compares manual and automatic labelling approaches, and assesses the reliability evidence reported for four resources. It distinguishes neutral labels that result from missed sentiment cues from genuine differences in interpretation. MESocSentiment illustrates the concern: 78.23% of its posts are labelled neutral, and its reported error examples include Malay expressions whose positive sentiment was missed. These examples warrant further validation but do not establish the corpus-wide error rate. The review also explains why class imbalance complicates the interpretation of Cohen's kappa. Future work should report agreement for the intended task, preserve individual judgements, develop guidance for Malaysian discourse particles such as lah, and test whether automatic annotators share errors. Improving label quality and documenting its limits are necessary steps towards more dependable training and evaluation.

Keywords: *sentiment analysis, code-mixed text, Manglish, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is the automated task of deciding whether a piece of text expresses a positive, negative or neutral opinion (Poria et al., 2023). Organisations use it to measure public opinion at scale, in domains that include consumer products and services, healthcare, political elections and financial services (Nazir et al., 2022). However, many tools used for this task assume that their input is written in one standard language. Joshi et al. (2020) highlight that only a small number of the world's languages are represented in current language technologies, with lexicons, pre-trained models and benchmark corpora concentrated in those few languages. Among these three, a pre-trained model is one that has already learned general patterns from a large body of text, and is then adapted to a task such as sentiment analysis (Devlin et al., 2019). Many of these language resources are developed primarily for individual languages, so text in which two languages alternate within a single message does not meet the conditions for which they were designed. Doğruöz et al. (2021) call this alternation code-switching, and report that its effect on model performance varies by task. In

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their survey, current models perform well on code-switched language identification and part-of-speech tagging, but only slightly better than chance on natural language inference. Winata et al. (2023), surveying a wider set of studies, also report that performance on code-switched text is often lower than performance on monolingual text for the same task, though not uniformly so. Code-switching between English and Malay is common in Malaysian social media text (AlAfnan et al., 2025).

Malaysia is a multilingual and multi-ethnic country. Drawing on Ethnologue figures, Coluzzi (2017) records around 140 languages in use in the country, and reports that code-mixing between a speaker's first languages and English or Malay is widespread in everyday communication. AlAfnan et al. (2025) report that English and Malay are among the languages most frequently combined in Malaysian written communication. The same study describes social media users alternating between the two languages within a single message rather than writing in one language at a time. Two local terms are used for this code-mixed writing, and they refer to different combinations of languages. AlAfnan et al. (2025) describe *Manglish* as a hybrid form that combines Malay and English, whereas Romadhona et al. (2022) describe *Bahasa Rojak* as a variety that combines English, Malay and Chinese. This review is concerned with Malay-English mixing. *Manglish* names that combination, while *Bahasa Rojak* names a broader one that also includes Chinese.

Manglish is not always described in the same terms. AlAfnan et al. (2025) note that it is sometimes viewed as having a negative effect on standard English and Malay rather than as a legitimate language variety in its own right. The structure of *Manglish*, however, is systematic rather than random. Treffers-Daller et al. (2022) analyse the classroom speech of two Malaysian teachers of English and find that Malay-English switching there follows describable syntactic patterns. AlAfnan et al. (2025) report that speakers use *Manglish* for identifiable purposes, which include communicating efficiently, marking identity and building rapport.

The same systematicity appears at the level of spelling. Maskat and Rahman (2020) document the Malay negator *tidak* being written as *x*, and *tidak ada* being written as *takde*, *xde* or *tade*. They sort this variation into five categories and map two of them back to their standard forms, and this categorisation suggests that the variation follows recurring patterns rather than occurring arbitrarily. A dictionary of standard Malay, such as *Kamus Dewan*, does not list these non-standard forms (Dewan Bahasa dan Pustaka, 2005). Tools that rely on such a dictionary may therefore treat them as out-of-vocabulary items, meaning words absent from the reference lexicon (Maskat and Rahman, 2020). Abu Bakar et al. (2020) describe Malay social media text as misspelled, out-of-vocabulary, ill-formed and noisy. This review reads that description as a statement about the reference standard being applied rather than about the care taken by the writer. These writing conventions are consistent within the observed variation, but they differ from the conventions that standard natural language processing tools were designed to handle.

These differences between Malaysian writing conventions and the assumptions built into standard tools raise a question about the source of the difficulty. When a sentiment model performs poorly on Malay-English mixed text, one common explanation concerns the model architecture (Doğruöz et al., 2021; Winata et al., 2023). Under this view the model processes code-switched input less effectively

than monolingual input, so a larger model, or one further trained on Malay-English or other Southeast Asian data, may perform better. The evidence for this explanation is mixed. Khanuja et al. (2020) report that multilingual models fine-tuned on code-switched data achieve the strongest scores among the systems evaluated on their benchmark. Adilazuarda et al. (2022) find that robustness to code-mixing varies with the embedded language, and attribute the variation to bias in the pre-training corpus. In contrast, Winata et al. (2021) find that pre-trained multilingual models do not necessarily produce high-quality representations of code-switched text. Conneau et al. (2020) describe a trade-off between two effects in multilingual models. Adding languages can improve performance on each of them, an effect they call positive transfer. Beyond a point, however, the modelling capacity available to any individual language falls, an effect they call capacity dilution, and the quality of its representations may decline. Doğruöz et al. (2021) identify a further limitation that is not architectural: the multilingual models in common use are trained mainly on monolingual corpora and may receive little code-switched text during pre-training. Model choice therefore accounts for part of the difficulty but not all of it.

Another source of difficulty arises one step earlier, at the point where sentiment labels are created. Supervised sentiment analysis requires a sentiment label for every message before a model can be trained or evaluated on it. The amount of labelled data available therefore sets a bound on what can be trained and on what can be tested. Poria et al. (2023) describe this reliance on annotated data as often limiting, and note that it has renewed interest in supplying models with external knowledge such as sentiment lexicons. A sentiment label is assigned either by a human annotator or by an automatic labelling method, and this step is called annotation. The labels form the supervision signal used to train a model, and the labels in a test set are the standard against which the model's predictions are compared. Incorrect labels can affect both model training and subsequent evaluation. During training, a model may learn the patterns associated with incorrect labels, which can reduce predictive accuracy (Frénay and Verleysen, 2014). During evaluation, incorrect labels in a test set can bias the measurement of model performance. Northcutt et al. (2021) show that label errors can change which of two models is ranked higher on a benchmark. Bowman and Dahl (2021) name annotation reliability as one of the conditions a sound evaluation benchmark must meet. An incorrect label can therefore have an effect at two separate stages, first when the model is trained and again when its performance is measured.

For Malaysian Malay-English code-mixed sentiment data, three findings make annotation reliability a priority. First, this review identifies four Malay-English sentiment resources (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024). None of them meets all four of the criteria a general-purpose resource would need. Those criteria are (1) at least ten thousand labelled messages; (2) a label distribution that is not dominated by a single sentiment class; (3) text drawn from everyday social media posts rather than from blogs or reviews; and (4) an assessment of annotation reliability by human annotators. Each of them satisfies some of these criteria but not all of them. Second, the largest, MESocSentiment, is labelled 78.23% neutral, and its reported error examples suggest that the labelling tools miss some Malay sentiment expressions (Mohd Shamsuddin et al., 2024). These examples do not establish how much of the

neutral class is incorrect. Third, two studies report no inter-annotator agreement coefficient (Kong et al., 2023; Mohd Shamsuddin et al., 2024). The other two report agreement for binary tasks, which does not establish reliability for positive, negative and neutral labelling (Kasmuri and Basiron, 2019; Romadhona et al., 2022).

The second and third findings are related. Reliability is normally assessed by having independent annotators label the same text and computing a chance-corrected agreement coefficient, such as Cohen's kappa (κ) (Cohen, 1960; Artstein and Poesio, 2008). Cohen's κ becomes difficult to interpret when one category dominates the annotators' label distributions, because expected chance agreement can rise with that skew (Feinstein and Cicchetti, 1990; Byrt et al., 1993). Di Eugenio and Glass (2004) call this effect the prevalence problem. A neutral-heavy corpus therefore needs both label validation and careful interpretation of agreement. These reliability concerns add to the shortage of annotated resources for code-mixed Malay text reported by Selvaraju et al. (2024).

These three findings lead to the question this review examines: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data have not yet been established. Linguistic description, model architecture and application domain are discussed only where they are relevant to that question. Two related topics are not examined here. The first is comparative benchmarking of sentiment classifiers, and the second is sentiment analysis as a general field. Both have already been surveyed in detail (Poria et al., 2023; Nazir et al., 2022). This review does not propose, implement or evaluate a system, and it does not report the authors' own experimental work. Its contribution is an analysis of why that question remains open, together with a set of directions for future research.

This review makes three contributions:

1. It compares manual annotation, labelling with existing sentiment tools, and generative-model annotation, examining their demands on human effort and their evidence of reliability.
2. It distinguishes missed sentiment cues from genuine differences in interpretation, and examines the reliability evidence reported for four relevant resources.
3. It explains how linguistic difficulties, labelling choices and class imbalance interact, and identifies research priorities for more reliable annotation and evaluation.

The literature examined here was identified through searches of the ACL Anthology, IEEE Xplore, Scopus and arXiv, combining terms for the language variety, the sentiment task and annotation quality. Studies that only benchmark classifiers, with no annotation or reliability component, were not included.

Section 2 examines linguistic challenges for annotation. Section 3 compares labelling approaches and their limitations. Section 4 assesses reliability evidence. Section 5 explains how these challenges interact and proposes research priorities, and Section 6 concludes.

2. Linguistic Challenges for Annotation

Reliable annotation requires attention to three features of Malay-English text: language switching, context-dependent particles and spelling variation.

2.1 Sentiment across language boundaries

A sentiment label must account for the whole message, even when its words come from different languages. Treffers-Daller et al. (2022) found frequent switching within clauses in Malaysian classroom speech. Their findings explain how Malay and English can combine within a sentence, although they do not establish how often this occurs on social media.

For annotation, the important question is whether the reader or tool recognises the words that express the opinion. A tool may understand the English part of a post but miss sentiment expressed in Malay. Fluent bilingual annotators can interpret both, but still need guidance on which opinion or target determines the label when a message contains several (Mohammad, 2016).

2.2 Particles and context

Particles such as *lah* can alter a message's tone without changing its factual content. Treffers-Daller et al. (2022) describe uses ranging from assertiveness to encouragement. AlAfnan et al. (2025) give "Don't disturb la, he's busy" as an example of a softened instruction. These studies draw on classroom speech and television dialogue, so they establish possible functions rather than their frequency in social media writing.

A particle does not have one fixed sentiment label. The instruction above might sound irritated or polite, depending on context. Guidelines therefore need to distinguish tone from positive or negative sentiment. Assigning *lah* a fixed polarity would remove the contextual distinction that makes it useful to speakers.

2.3 Spelling variation

Non-standard spelling can hide sentiment cues from automatic tools. Maskat and Rahman (2020) document forms such as *jgn* for *jangan* and several spellings of *macam mana*. Their spelling-normalisation experiment achieved a highest precision of 0.43: fewer than half of the proposed matches were correct. They removed English words and excluded code-mixed posts from the evaluation subsets, so this result cannot be treated as performance on Malay-English text.

The practical concern is especially clear for negation. As discussed in Section 1, *x* can represent *tidak*. Missing that form can change how an opinion is interpreted. Normalisation should therefore be checked for its effect on sentiment, rather than judged only by whether it produces standard-looking words.

3. Annotation Approaches and Their Limitations

3.1 Human and automatic labelling

These linguistic challenges affect both human and automatic labelling. Manual annotation allows readers to interpret local language use, but requires time and

relevant language skills. Kasmuri and Basiron (2019) developed a subjectivity corpus from blogs. Kong et al. (2023) used three independent annotators and a written guideline to label 11,568 tweets. Human involvement makes contextual judgement possible; it does not by itself establish reliability.

Human and automatic methods can also appear within one resource. SentiBahasaRojak combines automatically generated code-mixed reviews with stock-market posts labelled by five experts (Romadhona et al., 2022). Its reported agreement applies to the stock subset. This makes it important to describe how each part of a dataset was produced.

Existing sentiment tools can assign labels automatically. Mohd Shamsuddin et al. (2024) used TextBlob, VADER and pysentimiento to label MESocSentiment by majority vote. Although the study calls the construction process semi-automatic, its sentiment labels were assigned automatically. Whether those labels capture Malay-English sentiment must be checked separately.

Generative language models can also assign labels, but their ability to handle informal language needs task-specific validation. Jadhav et al. (2025) found that GPT-4o and Llama 3.1 405B underperformed a fine-tuned BERT model across four Marathi classification tasks, including sentiment analysis. This result does not establish their performance on Malaysian text. No study examined in this review directly evaluates generative annotation for Malay-English sentiment.

3.2 Interpreting neutral labels

MESocSentiment contains 19,714 posts, of which 78.23% are labelled neutral, 16.33% positive and 5.44% negative (Mohd Shamsuddin et al., 2024). A large neutral class is not itself evidence of error. It may reflect the sampled topics, the annotation rules, or genuine neutrality. The concern is whether some opinions received a neutral label because the tools missed their sentiment cues.

Two observations support checking this. The corpus was collected under the #Malaysia keyword across topics that include sports, politics and the arts (Mohd Shamsuddin et al., 2024). A comparable corpus of Malaysian tweets labelled by human annotators is 23.6% neutral (Kong et al., 2023). Neither observation shows that the neutral labels are wrong, but together they justify validation.

Lexicon-based tools are vulnerable to this problem when an expression is absent from their resources. VADER combines recognised sentiment cues into a score and assigns neutral to scores near zero (Hutto and Gilbert, 2014). If a Malay opinion contributes no recognised cue, a neutral output may reflect a coverage gap.

Mohd Shamsuddin et al. (2024) present six labelled examples, three of which misclassify positive Malay expressions as neutral. One is *memang cantik sangat*, meaning “really very beautiful”. These selected cases support checking for missed Malay sentiment, but they cannot estimate the error rate across the corpus.

Disagreement can reflect plausible differences in interpretation rather than annotation error (Plank, 2022). Pavlick and Kwiatkowski (2019) show that disagreement can persist in textual inference even when more judgements are

collected. Neither finding establishes that ambiguity caused the neutral skew in MESocSentiment. A neutral label should not combine “no sentiment” with “uncertain interpretation” without distinguishing the two.

3.3 Limits of majority voting

Majority voting gives every annotator the same weight (Hovy et al., 2013), so an error shared by several annotators is retained. Table 1 shows that all three MESocSentiment analysers assigned neutral to most retained posts. The analyser with the highest neutral share, pysentimiento, is the only one of the three built on a pre-trained transformer model (Devlin et al., 2019), so greater model capacity did not reduce the skew. However, class totals alone do not establish whether the tools mislabelled the same posts. That requires comparison against labels checked independently by bilingual reviewers.

Table 1. Neutral labels in the retained MESocSentiment corpus

Analyser	Neutral posts	Share
TextBlob	14,305	72.56%
VADER	13,115	66.53%
pysentimiento	16,916	85.81%
Majority vote	15,422	78.23%

Source: Mohd Shamsuddin et al. (2024). All shares use the 19,714 retained posts as the denominator.

Mohd Shamsuddin et al. (2024) discarded 512 of 20,226 posts because each analyser assigned a different label. Removing them left the neutral count unchanged at 15,422 while reducing the total from 20,226 to 19,714, so the neutral share rose by about two percentage points. Removing disputed posts may also exclude difficult cases, but the published counts do not establish their linguistic characteristics. Retaining their text and individual labels would allow further analysis of disagreement, following the approach advocated by Uma et al. (2021).

4. Evaluating Annotation Reliability

Shared errors limit what agreement can establish: consistent judgements do not necessarily indicate correct labels. Cohen’s κ compares observed agreement between two annotators with agreement expected from their label distributions (Cohen, 1960). In computational linguistics, values above 0.8 are generally treated as good reliability, and values between 0.67 and 0.8 as allowing tentative conclusions (Artstein and Poesio, 2008). The same survey cautions that interpreting a value requires knowing the task, the annotators and the distribution of their labels.

Class imbalance can make κ difficult to interpret. Di Eugenio and Glass (2004) illustrate this with two binary annotation examples: both have 90% observed agreement, yet κ ranges from −0.048 to 0.80 because the category distributions differ. A single threshold therefore cannot be applied to every agreement result.

For a neutral-heavy dataset, researchers should examine which classes attract agreement and which attract disagreement. The final corpus distribution is a reason to investigate prevalence, but κ depends on the individual annotators' distributions. Reliability assessment remains possible; it needs more evidence than an aggregate coefficient or class count.

Table 2. Reliability evidence reported for the reviewed resources

Resource	Annotation evidence	Scope and limitation
MY-EN-CS Kasmuri and Basiron (2019)	Two bilingual undergraduates; written guideline; κ = 0.83.	Opinionated versus factual judgement. Does not establish three-class polarity agreement.
MyCovid-Senti Kong et al. (2023)	Three independent annotators; written guideline.	Three-class sentiment labels; no agreement coefficient reported.
SentiBahasaRojak Romadhona et al. (2022)	Five experts labelled stock posts; reported kappa = 0.77.	Positive versus negative stock labels; coefficient variant unspecified. Other subsets were generated automatically.
MESocSentiment Mohd Shamsuddin et al. (2024)	Majority vote among three automatic analysers.	No human inter-annotator agreement coefficient reported.

Table 2 shows gaps in the available evidence, rather than proving that all four resources are unreliable. Two studies report no coefficient; the other two report binary-task agreement. Those binary results are informative for their stated tasks, but cannot be transferred to three-class sentiment annotation. Romadhona et al. (2022), for example, used five experts with the final label set by majority vote, whereas Cohen's κ is defined for two annotators. Studies with more than two annotators should therefore name the coefficient used and explain how it was calculated.

5. Research Priorities

The reviewed studies leave uncertainty about how labels were produced and checked. Section 5.1 first sets out how the challenges described in Sections 2 to 4 interact. Four research priorities then follow: annotation reporting, agreement under class imbalance, guidance for particles and ambiguity, and testing for shared errors.

5.1 How the challenges interact

The three challenges examined above are connected. Particles and spelling variation mean that some sentiment cues can be read in more than one way or missed altogether. Reliable labelling therefore needs fluent bilingual readers, and their cost is the reason Mohd Shamsuddin et al. (2024) give for automating. The tools used for automation were designed for other languages. Where none of them recognises a Malay sentiment cue, majority voting confirms the missed cue rather than correcting it, and the neutral class grows.

That skew then affects the check that would reveal it. A label distribution dominated by one class is the condition under which Cohen's κ becomes difficult

to interpret (Feinstein and Cicchetti, 1990; Byrt et al., 1993). A researcher who suspects the labels and computes κ therefore cannot easily distinguish poor annotation from competent annotation of a skewed corpus. Corpora whose reliability was never established are then released and used to train and evaluate models. A model trained on such labels learns the patterns associated with them (Frénay and Verleysen, 2014). If that model is later used as an annotator, the sequence repeats.

This sequence explains why competent work at each stage has not accumulated into progress. Each reviewed study addresses one stage, but an intervention at one stage is assessed with a statistic that another stage has made difficult to interpret. The four priorities below each address one point in the sequence.

5.2 Report how labels were produced

Corpus releases should state who or what assigned the labels, provide the guideline, and identify the task and agreement measure. Data statements can document the populations and conditions to which a resource applies (Bender and Friedman, 2018). Releases should also report class distributions, agreement by class, individual judgements and discarded items. Preserving individual annotations supports methods that model differences between annotators instead of reducing them to a majority label (Mostafazadeh Davani et al., 2022).

5.3 Evaluate agreement under class imbalance

Researchers should compare agreement measures on data that retain independent annotations, examining how results change as class proportions vary. One option is prevalence adjustment, which changes the chance-agreement calculation to reduce the effect of unequal category frequencies (Byrt et al., 1993). Its assumptions and suitability for three-class sentiment need evaluation; adjustment does not automatically make a measure appropriate (Di Eugenio and Glass, 2004). Corpora containing only final labels would require additional annotation.

5.4 Develop guidance for particles and ambiguity

Annotation guidelines should show how context changes the interpretation of particles such as *lah* and *kan*. Mohammad (2016) separates questions about the speaker, opinion target and attitude, and Kong et al. (2023) adapt this approach for Malaysian tweets. A useful extension would include local particle examples and a clear rule for unresolved ambiguity. Shared guidance can reduce avoidable disagreement while preserving readings that remain genuinely uncertain.

5.5 Test whether annotators share errors

Bilingual humans, existing sentiment tools and generative models should label the same Malay-English sample. A separate group of bilingual reviewers could assign reference labels independently, explain their decisions and discuss disagreements. Unresolved cases should retain the competing labels. Human review of this kind can also improve the automatic detection of annotation errors (Weber and Plank, 2023). Weber-Genzel et al. (2024) use explanations and

subsequent validity judgements to distinguish annotation error from plausible variation in English textual inference. A similar procedure could support the proposed Malay-English comparison. Researchers could then test whether voting reduces errors or preserves mistakes shared by the annotators.

6. Conclusion

This review examined why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data have not yet been established. Part of the difficulty lies in the models, but another part arises earlier, in how sentiment labels are produced and checked. The reviewed resources show incomplete agreement reporting, evidence of missed Malay sentiment cues, and uncertainty about the benefits of majority voting. These difficulties are connected. The labelling methods produce skewed class distributions, and that skew makes the agreement coefficient used to check the labels difficult to interpret. Progress therefore depends on transparent annotation reporting, guidance for context-sensitive expressions, agreement analysis that accounts for class imbalance, and direct tests of whether annotators share errors. Until reliability can be assessed under these conditions, a new corpus cannot be shown to improve on the ones examined here.

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